CS 401: Systems Programming - Course Overview

Term: Fall 2026 | Credits: 4.0 | Location: Engineering Building Room 101.

Instructor: Course Staff (contact via course portal). Office Hours: Mondays and Wednesdays from 2:00 PM to 4:00 PM in Computer Science Building Room 204. Teaching Assistant Office Hours: Fridays from 10:00 AM to 12:00 PM online via course discussion forum.

Prerequisites: Students must have completed CS 201 (Data Structures and Algorithms) and CS 202 (Computer Architecture and Systems) with a grade of C or better. Proficiency in C/C++ programming and Unix command-line utilities is required.

Course Description: This course covers fundamental and advanced principles of systems programming, concurrency, memory management, and distributed systems. Students develop multi-threaded network services and high-performance storage components.

Grading Breakdown and Assessment Policies

Grading Breakdown: Course grades are computed from the following components (totaling exactly 100 percent):

- Programming Projects (4 assignments): 40 percent of total grade (10 percent each).

- Midterm Examination: 25 percent of total grade.

- Final Term Examination: 25 percent of total grade.

- Homework Problem Sets and Quizzes: 10 percent of total grade.

Grading Scale: A: 90-100 percent, B: 80-89 percent, C: 70-79 percent, D: 60-69 percent, F: below 60 percent.

Late Policy: Assignments submitted after the scheduled deadline will incur a penalty of 10 percent per day for up to 3 days late. Submissions more than 3 days late will receive zero credit. Medical and emergency accommodations require prior approval from the academic dean.

Examination Schedule and Midterm Details

Midterm Examination Date and Weight: The Midterm Examination is scheduled for Thursday, November 5, 2026, from 7:00 PM to 9:00 PM in Lecture Hall C. The midterm accounts for 25 percent of the overall course grade.

Midterm Scope and Allowed Materials: The examination covers all material from Weeks 1 through 7, including memory hierarchies, thread synchronization, and asynchronous I/O. The exam is closed-book; students may bring one double-sided letter-sized (8.5x11 inch) handwritten formula sheet.

Final Examination: The Final Examination will be held during university exam week on December 18, 2026, contributing 25 percent of the course grade. It is comprehensive with an emphasis on Weeks 8 through 14.

Weekly Course Schedule and Topics

Course Topics by Week:

- Week 1: Unix Environment, Tooling, Compilers, and Build Systems.
- Week 2: Memory Layout, Pointers, Stack and Heap Dynamic Allocation.
- Week 3: POSIX File I/O, File Descriptors, and System Calls.
- Week 4: Processes, Signals, IPC, and Process Management.
- Week 5: Threads, Pthreads Library, and Shared Memory.
- Week 6: Concurrency, Mutexes, Semaphores, and Race Condition Prevention.
- Week 7: Deadlocks, Condition Variables, and Thread Synchronization.
- Week 8: Midterm Examination Review and Midterm Exam.
- Week 9: Network Sockets, TCP/IP Communication, and Client-Server Protocols.
- Week 10: Asynchronous and Event-Driven I/O Architectures (epoll/kqueue).
- Week 11: Distributed Consensus, RPC Systems, and Fault Tolerance.
- Week 12: File System Design, Disk Layouts, and Crash Recovery.
- Week 13: Memory-Mapped Files, Caching Strategies, and Performance Profiling.
- Week 14: Final Review, Case Studies in Modern Operating Systems, and Project Presentations.